TAYLOR'S UNIVERSITY
Wisdom · Integrity · Excellence

SCHOOL OF COMPUTER SCIENCE

ITS61504 Data Mining

GROUP ASSIGNMENT-3

HAND OUT DATE AND TIME: July 22, 2026, 11:59 PM

Instructions to students:

- The assignment should be attempted in groups.
- Complete this cover sheet and attach it to your submission – this should be your first page.

Module Learning Outcome:
MLO3 - Work collaboratively to design, execute, and complete a predictive modelling project, demonstrating accountability, autonomy, and ethical responsibility

Student declaration:	
I declare that: ▪ ***I understand what is meant by plagiarism.*** ▪ ***The implication of plagiarism and usage of AI generative tool have been explained to us by our lecturer.*** ***This project is all our work, and I have acknowledged any use of the published or unpublished works of other people.***	
NAME **(change accordingly for group or individual assignment)**	
Name	**Student ID**
1. Group Leader's Name	

Avoid copy and paste job in your report and it is considered as plagiarism. Plagiarism in all forms is forbidden. Students who submit plagiarized assignment will deserve 0 marks.

Assessment Criteria

Assessment Task	Weightage	MLO Assessed	Assessment Instrument	Topics	Week	MQF 2.0
Group Project MLO3	40%	1	N/A	1-12	14	3f

MLO3: Work collaboratively to design, execute, and complete a predictive modelling project, demonstrating accountability, autonomy, and ethical responsibility

C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, C3B = Interpersonal Skills, **C3C** = Communication Skills, **C3D** = Digital Skills, **C3E =** Numeracy Skills**, C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Entrepreneurial Skills, **C5** = Ethics & Professionalism

Part-I: Assignment Submission Requirements (TOTAL: 40 Marks)

1. Project Report (Softcopy in PDF and DOC)
a. Cover page
b. Marking Rubric
c. A clear work breakdown structure to describe what each member is doing
d. Report Writing Guidelines:

The report should be structured in a coherent and scholarly manner, covering the following key sections:

Question 1: Introduction and Purpose of Study (4 Marks)

a) Provide the general context of the study/project. (1 Marks)
b) Clearly define the problem statement and explain how/why data mining is used. (2 Marks)
c) Describe the business or research motivation and expected outcomes. (1 Marks)

Question 2: Description of Data (4 Marks)
a) Describe the source of the dataset. (1 Marks)
b) Explain the type and structure of the data. (1 Marks)
c) Identify key attributes/features of the dataset. (1 Marks)
d) Present a representative sample (up to 1 page) to illustrate data characteristics.(1 Marks)

Question 3: Data Problems and Pre-Processing (4 Marks)
a) Identify issues in the dataset (e.g., missing, noisy, inconsistent data). (1 Marks)
b) Explain the preprocessing techniques applied (e.g., cleaning, transformation, normalization). (2 Marks)
c) Provide before-and-after comparisons to demonstrate improvements. (1 Marks)

Question 4: Data Mining Techniques (4 Marks)
a) Identify the data mining technique(s) used (e.g., Apriori, classification, clustering). (1 Marks)
b) Justify why the technique(s) were selected for the problem and dataset. (1 Marks)
c) Briefly describe how the technique(s) were implemented. (2 Marks)

Question 5: Model Validation (4 Marks)
a) Describe how the model was evaluated. (1 Marks)
b) Identify the metrics used (e.g., accuracy, support, confidence, lift). (1 Marks)
c) Explain validation techniques or experimental design used to ensure reliability. (2 Marks)

Question 6: Conclusion and Future Improvements (3 Marks)
a) Summarize key findings and insights from the analysis. (1 Marks)
b) Discuss the significance of patterns, trends, and relationships identified. (1 Marks)
c) Highlight limitations and suggest improvements or future research directions. (1 Marks)

Question 7: Python Implementation (2 Marks)
a) Provide well-structured and documented Python code for preprocessing and analysis. (1 Marks)
b) Ensure reproducibility and clarity of implementation. (1 Marks)

Question 8: References (1 Mark)
a) List all references, tools, libraries, and academic sources used. (0.5 Marks)
b) Follow an appropriate citation style (e.g., APA). (0.5 Marks)

Question 9: Project Presentation and Viva (14 Marks)
a) Clarity and organization of presentation (4 Marks)
b) Understanding of project and methodology (4 Marks)
c) Explanation of results and ability to answer questions (4 Marks)
d) Team participation and communication will directly affect individual/group marks. (2 Marks)

Ideas for Projects / Reference

You can refer to this site for some project ideas: -

https://archive.ics.uci.edu/ml/datasets.html But you can get your data from any other

suitable sites. Data sets: https://www.kaggle.com/

** Few more data sets are also used:

1. https://www.kaggle.com/code/onesinustamba/deep-learning-cnn-dcnn-for-brain-tumor-detection
2. https://www.kaggle.com/datasets/pkdarabi/medical-image-dataset-brain-tumor-detection/code
3. https://www.kaggle.com/code/saife245/melanoma-detail-analysis-eda-ip-augmentation-model
4. https://www.kaggle.com/code/sasakitetsuya/smoker-lung-cancer-stage-detection-model
5. https://www.kaggle.com/code/itsuki9180/mvan-covid-mrna-vaccine-analysis-notebook-268
6. https://www.kaggle.com/code/allunia/pulmonary-dicom-preprocessing

(Hint: Convert images data sets in CSV file using library of CSV)

- ***Students can use MONAI datasets and any publically available datasets also.***

Part-II Submission Requirements

1. Report Formatting:

- Font Type: Times New Roman
- Font Size: 12
- Line Spacing: 1.5
- Alignment: Justified

2. Report Structure:

- Page Limit: 5 – 25 pages (strictly adhere to this range).

- Cover Page: Must include Name, ID, Date, Signature, and Score.
- Marking Rubrics: Attach the grading rubric as the second page.
- Source Code: Include the full implementation of your sentiment analysis approach.
- Answer Script Report: Provide a structured report summarizing your analysis and findings.

Notes:

- Emphasize practical application.
- Brief explanations for each step.
- Use in-text citations and provide a list of references at the end (APA format).
- The Turnitin similarity score should be below 15% overall and less than 1% from a single source, excluding program source codes.

The following items are to be handed in:

- Submission of softcopy version to MyTIMeS
- Soft copy version of the *project report in PDF* and *DOC* version must be uploaded into My TiMES before your presentation schedule. Failed to do so, 30% will be deducted from your project

Question / Criteria	Marks	Excellent (Full Marks)	Good	Satisfactory	Needs Improvement / Poor
Introduction & Purpose of Study	4	Clear, concise context; well-defined problem; strong justification for data mining use and expected outcomes.	Good explanation with minor gaps in clarity or justification.	Basic description; problem or purpose not clearly articulated.	Unclear or missing context/problem; no justification.
Description of Data	4	Data source clearly identified; structure and attributes well explained; relevant sample provided.	Source and structure explained with minor missing details.	General dataset description; limited attribute explanation or sample.	Poor or missing dataset description.
Data Problems & Preprocessing	4	All major data issues identified; thorough preprocessing with clear before/after comparison.	Most issues identified; preprocessing applied with some explanation.	Limited discussion of issues; basic preprocessing shown.	No clear preprocessing or issues identified.
Data Mining Techniques	4	Appropriate technique clearly explained and strongly justified; implementation well described.	Technique explained with reasonable justification; minor gaps.	Basic explanation with weak justification.	Incorrect or unclear technique selection.
Model Validation	4	Clear validation approach; appropriate metrics (e.g., accuracy, support,	Validation explained with some metrics; minor clarity issues.	Basic validation with limited metric explanation.	No or incorrect validation approach.

		confidence, lift) well explained.			
Conclusion & Future Improvements	3	Strong conclusions; meaningful insights; clear limitations and realistic future improvements.	Good conclusions with some insights and improvements.	General conclusions; limited depth or unclear improvements.	Weak or missing conclusions and future direction.
Python Implementation	2	Well-structured, clean, and fully documented code; reproducible results.	Code is functional and mostly clear; minor documentation issues.	Code provided but lacks clarity or structure.	Missing, incomplete, or non-functional code.
References	1	Complete and correctly formatted references (APA or appropriate style).	Minor formatting issues in references.	Incomplete references or inconsistent style.	Missing or incorrect references.

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Assignment 3 Marking Rubrics

Criteria	Weightage	Beginning (0–49%)	Developing (50–64%)	Mastering (65–79%)	Excellent (80–100%)
Work collaboratively to design, execute, and complete a predictive modelling project, demonstrating accountability, autonomy, and ethical responsibility	10.00%	Demonstrates limited clarity, organization, and comprehensiveness. Objectives are unclear or poorly defined. Literature review is insufficient or irrelevant. Data description lacks detail. Methodology is weak or unjustified. Project plan is unrealistic or missing. Expected outcomes and impact are unclear. Challenges and limitations are not adequately addressed. Team roles and responsibilities are poorly defined.	Demonstrates moderate clarity and organization. Objectives are partially defined. Literature review is somewhat relevant. Data description is basic. Methodology is somewhat justified. Project plan is present but lacks detail. Expected outcomes and impact are somewhat described. Some challenges are identified. Team roles are partially defined.	Demonstrates good clarity, organization, and comprehensiveness. Objectives are well-defined and justified. Literature review is relevant. Data description is clear. Methodology is justified and feasible. Project plan is realistic. Expected outcomes and impact are clearly described. Challenges and limitations are addressed. Team roles and responsibilities are defined.	Demonstrates an exceptional level of clarity, organization, and comprehensiveness. Objectives are well-defined and strongly justified. Literature review is thorough and highly relevant. Data description is detailed and clear. Methodology is well-justified and feasible. Project plan and timeline are highly realistic and detailed. Expected outcomes and impact are clearly articulated. Challenges and limitations are critically addressed. Team roles and responsibilities are clearly and effectively defined.
Project Implementation, Data Processing, and Results Presentation	30.00%	Demonstrates limited execution and poor organization. Data preprocessing is inadequate or incorrect. Analysis is weak or incomplete. Results are unclear or poorly presented. Report lacks clarity and coherence.	Demonstrates basic execution with some organization. Data preprocessing is partially correct. Analysis is present but lacks depth. Results are somewhat clear. Report shows some structure but lacks consistency.	Demonstrates good execution and organization. Data preprocessing is appropriate. Analysis is meaningful. Results are clearly presented. Report is structured and understandable.	Demonstrates excellent execution and organization. Data preprocessing and model implementation are performed effectively. Results are clearly presented in a well-structured and coherent report. Insights are meaningful, with strong interpretation and clear community or business implications.

END OF ASSIGNMENT-3